

Performance Comparison of Quantum Error-Correcting Codes

Course: Classical and Quantum Error Correction

2026

Motivation & Goal

- ▶ QEC is essential for fault-tolerant quantum algorithms, since quantum computers are inherently noisy.
- ▶ Different QEC codes offer different trade-offs in terms of qubit overhead, performance, and code structure.
- ▶ Goal: Compare three codes under three noise models
 - ▶ 3-qubit repetition $[[3,1,1]]$
 - ▶ 5-qubit perfect $[[5,1,3]]$
 - ▶ Steane $[[7,1,3]]$
- ▶ Three noise models: bit-flip, phase-flip, depolarizing
- ▶ Main question: how do different code structures and resource overheads affect logical performance under different noise models?

Codes: Stabilizer structure

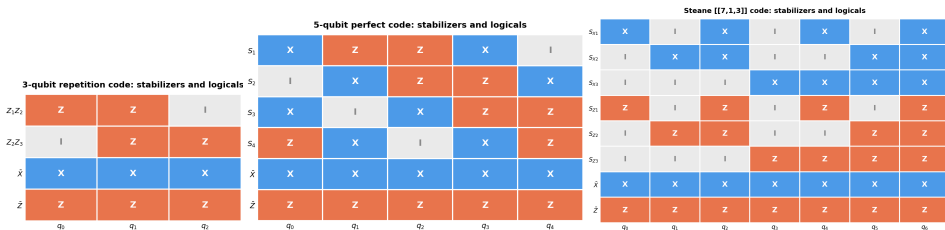


Figure: Visualization of the stabilizers and logical Pauli operators of the three codes.

- ▶ 3-qubit repetition code: CSS, two Z-type stabilizers
- ▶ 5-qubit perfect code: non-CSS, four mixed XZ-type stabilizers
- ▶ Steane $[[7,1,3]]$ code: CSS, three X-type stabilizers and three Z-type stabilizers

Noise models studied

Noise model	Error process
Bit-flip	X with probability p
Phase-flip	Z with probability p
Depolarizing	X, Y, Z with equal probability $\frac{p}{3}$

Assumption: independent, identical single-qubit errors with physical error probability p .

Method: Simulation pipeline



Method 1: full quantum circuit simulation in Qiskit - encoding circuits (generated using Clifford synthesis), errors implemented as Pauli gates, ancilla-based syndrome extraction, measurement

- ▶ Can be extended to more realistic noise models, but computationally slow → fewer trials and noisy results

Method 2: Monte Carlo simulation (using stabilizer formalism) to make a large parameter sweep over physical error probabilities → allows faster and smoother performance comparison plots

- ▶ Tracks stabilizers instead of the full statevector
- ▶ For every (code, noise model, p) triple, 50k-3M trials. 220 configurations in total.

Method: Decoding strategies

Same lookup tables for both methods:

- ▶ **3-qubit repetition code:** lookup-table decoding
 - ▶ 2 stabilizers $\rightarrow$ 4 syndromes $\rightarrow$ majority-vote correction on the flipped qubit.
- ▶ **5-qubit perfect code:** stabilizer-based decoding
 - ▶ Generic brute-force enumeration of all weight-1 Pauli errors on 5 qubits. 4 stabilizers $\rightarrow$ 16 syndromes, exactly matching the 15 non-trivial weight-1 Pauli errors + identity ("perfect" code).
- ▶ **Steane code:** lookup-table decoding
 - ▶ CSS from classical Hamming(7,4): the 3-bit syndrome read as a binary number gives the error location directly. X-errors and Z-errors decoded independently.

Results: Simulation validation

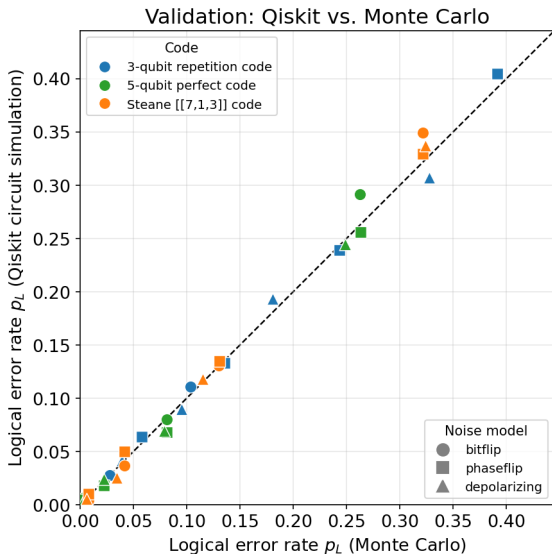


Figure: Qiskit circuit simulation p_L vs. Monte Carlo p_L . For Qiskit circuit simulation 36 configurations run, for $p \in \{0.02, 0.05, 0.1, 0.2\}$

Results: bit-flip noise

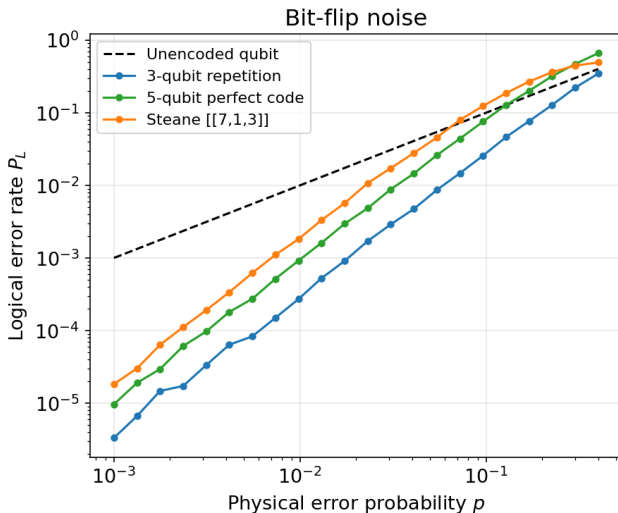


Figure: Logical error rate P_L against physical error probability p under bit-flip noise.

Results: phase-flip noise

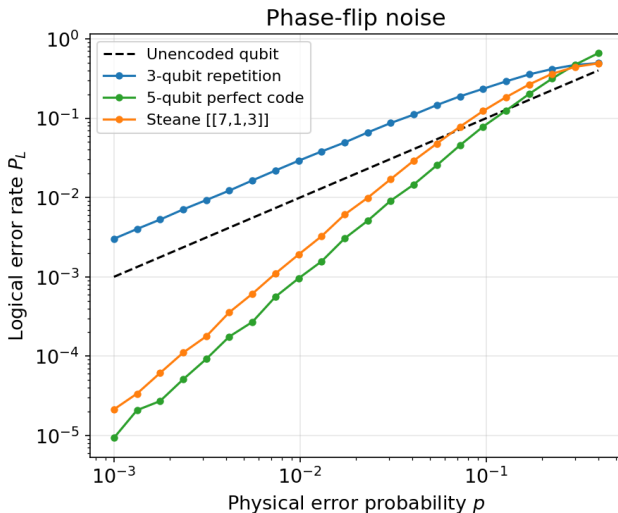


Figure: Logical error rate P_L against physical error probability p under phase-flip noise.

Results: depolarizing noise

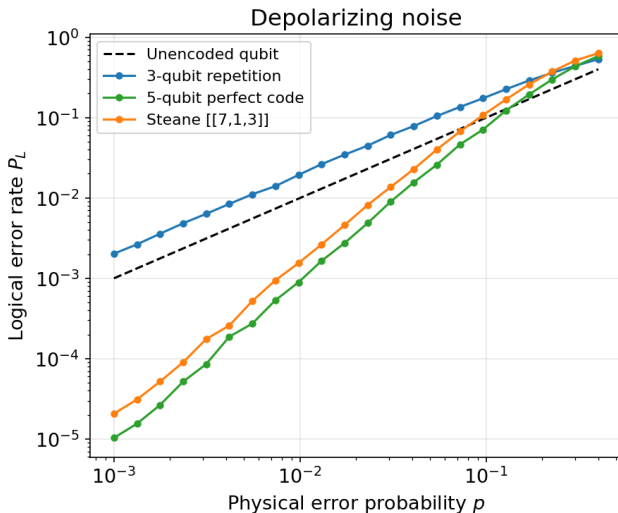


Figure: Logical error rate P_L against physical error probability p under depolarizing noise.

Results: Pseudo-thresholds

- ▶ The pseudo-threshold p^* is the physical error rate p where the code's logical error rate equals the unencoded physical error rate.
- ▶ For a physical error probability below the pseudo-threshold, $p < p^*$, encoding improves performance, above it, it worsens the performance.

	$p^*(\text{bit-flip})$	$p^*(\text{phase-flip})$	$p^*(\text{depolarizing})$
3-qubit repetition	≈ 0.50	-	-
5-qubit perfect code	≈ 0.128	≈ 0.131	≈ 0.133
Steane $[[7,1,3]]$ code	≈ 0.063	≈ 0.062	≈ 0.078

Discussion & Conclusions

- ▶ **Encoding needs to match the noise type:** the 3-qubit repetition code performs best for bit-flip noise but performs worse than the unencoded qubit for phase-flip and depolarizing noise.
- ▶ **More qubit overhead is not necessarily better:** the 5-qubit and Steane codes (both distance-3) perform similarly for all noise models, despite using 5 vs. 7 physical qubits. In fact, the 5-qubit perfect code performs better and reaches a higher pseudo-threshold than the 7-qubit Steane code, despite having fewer physical qubits.
- ▶ **Code structure matters:** although the 5-qubit perfect code performs well with low qubit overhead, its non-CSS structure makes implementation more complex, whereas the structure of CSS codes such as Steane offer advantages for implementation and scaling.

Limitations & Extensions

- ▶ Simplified noise model: one noise event after encoding, ideal syndrome extraction, no measurement errors
 - ▶ Possible extension: could add circuit-level noise (gate and measurement errors) through Qiskit NoiseModel
- ▶ Assumption of independent, identical single-qubit noise. No correlated noise.
 - ▶ Possible extension: investigating correlated noise
- ▶ Generic noise parameters: physical error rates are not hardware-specific.
 - ▶ Possible extension: could use hardware-calibrated noise parameters (from real superconducting QPUs).

References

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- ▶ Roffe, J., Quantum Error Correction: An Introductory Guide, Contemporary Physics, 60, 3 (2019)
- ▶ Steane, A. M., "Error Correcting Codes in Quantum Theory," Phys. Rev. Lett. 77, 793 (1996)